

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: Interventional platform study investigating the impact of digital health solutions on health outcomes and health-care resource utilization in participants receiving systemic treatment in clinical practice

Short Title/Acronym: ORIGAMA

Protocol Number: M042720 / 2023-504342-55-00

NCT Number: NCT05694013

Sponsor/Company Name: Roche (F. Hoffmann-La Roche Ltd)

Investigational Product: Digital Patient Monitoring (DPM) solution (Roche DPM Module / Kaiku Health) and Atezolizumab (TECENTRIQ)

Phase of Development: PHASE2, PHASE3

Trial Status: TERMINATED (Study was terminated early)

Medical Monitor: Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective: To evaluate the clinical impact and utility of digital health solutions (DHS), specifically the Roche DPM Module, on health outcomes, patient-reported symptom interference, and health-care resource utilization in people receiving systemic anti-cancer treatment (approved or non-approved) in clinical practice.

Secondary Objectives: To assess the feasibility of flexible care (at-home administration) adoption, safety, overall survival (exploratory), and other patient-reported outcomes (PROs).

2.2 Background & Rationale

Disease- and treatment-related symptoms negatively affect patients' quality of life and may become life-threatening. There is an unmet need for improved understanding, prevention, and mitigation of symptoms, as well as tools to facilitate remote tracking of patient-reported outcomes (PROs) to support self-management. As anticancer treatments are increasingly administered outside of traditional clinical settings, Digital Patient Monitoring (DPM) tools are critical to remote symptom management. This study investigates the real-world clinical utility and impact of DPM.

3. Study Design & Methodology

3.1 Design Framework

Study Type: Interventional Clinical Trial, open-label, multi-

country platform study.

Preferred UMLS Name: Clinical Trial

Control Type: Active Comparator (Standard of Care support without DPM in Cohort A).

Blinding: Open-label.

Randomization: 1:1 Randomization (for Cohort A).

3.2 Planned Enrollment & Duration

Targeted Enrollment: 49 participants (Note: Originally estimated higher, but revised to 49 prior to early termination).

Number of Sites: Multi-country clinical practice sites.

Estimated Study Start: February 2023.

Estimated Primary Completion: Upon completion of $\geq$ 12 weeks of follow-up for all randomized patients in Cohort A and 16 cycles of treatment for Cohort B.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

All Participants: * Email address, access to an internet-capable device (smartphone, tablet, or PC), and access to an internet connection.

Cohort A: * Histologically confirmed diagnosis for mNSCLC, ES-SCLC, or HCC (Child Pugh A). * Systemic therapy naive. * Prescribed an atezolizumab IV regimen. * Eastern Cooperative Oncology Group (ECOG) Performance Status of 0, 1, or 2.

Cohort B: * Complete resection of a histologically or cytologically confirmed Stage IIB-IIIB (T3-N2) NSCLC. * PD-L1 positive. * Have completed adjuvant chemotherapy at least 4 weeks and up to 12 weeks prior to randomization and must be adequately recovered from chemotherapy treatment. * ECOG Performance Status of 0 or 1. * Adequate hematologic and end-organ function. * For participants receiving therapeutic anticoagulation: stable anticoagulant regimen. * Negative for hepatitis B virus (HBV) or hepatitis C virus (HCV).

4.2 Exclusion Criteria

All Participants: * Any physical or cognitive condition that would prevent the participant from using the DHS. * Participants not proficient with any of the available DHS language translations or with psychiatric/neurologic disorders or any condition that may impact the participant's ability to use the DPM solution. * Currently participating in another interventional trial. * History of malignancy within 5 years prior to initiation of study treatment, with the exception of the cancer under investigation in this study and malignancies with a negligible risk of metastasis or death.

Cohort A: * Concomitant anti-cancer therapy at the time of starting atezolizumab (IV) regimen on the index date which is not part of a locally approved combination therapy with atezolizumab.

* Participants not receiving atezolizumab, but an atezolizumab biosimilar or non-comparable biologic. * Participants currently using another DPM or ePRO solution for symptom management and/or reporting.

Cohort B: * Participants known to have a sensitizing mutation in the EGFR gene or an ALK fusion oncogene. * Uncontrolled tumor-related pain. * Uncontrolled pleural effusion, pericardial effusion, or ascites requiring recurrent drainage procedures (once monthly or more frequently). * History of leptomeningeal disease or spinal cord compression not definitively treated/stable. * Uncontrolled or symptomatic hypercalcemia. * Active or history of autoimmune disease or immune deficiency. * History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, or idiopathic pneumonitis, or evidence of active pneumonitis on screening chest CT. * Active tuberculosis or significant cardiovascular disease. * Major surgical procedure within 4 weeks prior to initiation, or anticipation of need for one. * Severe infection within 4 weeks prior to initiation, or treatment with therapeutic oral/IV antibiotics within 2 weeks. * Prior allogeneic stem cell or solid organ transplantation. * Treatment with a live, attenuated vaccine within 4 weeks prior to treatment. * Current treatment with anti-viral therapy for HBV. * Treatment with investigational therapy within 28 days prior to initiation. * Prior treatment with CD137 agonists or immune checkpoint blockade therapies (anti-CTLA-4, anti-PD-1, anti-PD-L1). * Treatment with systemic immunostimulatory agents within 4 weeks or systemic immunosuppressive medication within 2 weeks prior to initiation. * History of severe allergic anaphylactic reactions, known hypersensitivity to CHO cell products, or atezolizumab formulation components (including hyaluronidase). * Pregnancy or breastfeeding. * Pathology that could interfere with protocol-specified outcome assessments. * Participants currently using another DPM or ePRO solution.

5. Intervention & Treatment Plan

5.1 Investigational Regimen

Drug Name / Preferred Name: atezolizumab

Trade Name: Tecentriq

ATC Code: L01FF05

EMA URL: https://www.ema.europa.eu/en/documents/product-information/tecentriq-epar-product-information_en.pdf

Dosage/Frequency:

Cohort A: Intravenous (IV) atezolizumab administered via locally approved anticancer regimens.

Cohort B: Subcutaneous (SC) atezolizumab (1875 mg; Day 1 of each 21-day cycle).

Route of Administration: Intravenous (IV) or Subcutaneous (SC) alongside the digital web/app-based DPM module.

Duration of Treatment: Cohort A participants receive treatment until progressive disease or per local standard-of-care. Cohort B

participants receive 3 hospital cycles followed by 13 cycles at home.

5.2 Concomitant & Prohibited Therapy

Permitted: Standard of care (SoC) supportive treatments.

Prohibited: Any therapies strictly conflicting with the atezolizumab systemic therapy protocol.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Prior to Tx | Informed Consent, Demographics, ECOG Status, App Setup | | **Baseline** | Day 0 | Randomization, First Dose Administration, PRO Baseline entry | | **Interim** | Q3W / Q6W | Regular clinical visits, Atezolizumab dosing. Q6W reporting on HRQoL and symptom impact | | **Primary Assessment** | Week 12 (Cohort A) | Primary evaluation of the Total Symptom Interference Score via DPM | | **End of Study** | Varies | Completion of 16 cycles (Cohort B) or disease progression (Cohort A) - *Note: Terminated early.* |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint

Definition:

*Cohort A: Mean Difference in Change of Week 12 Value From Baseline of Participant-reported Total Symptom Interference Score From the MD Anderson Symptom Inventory (MDASI) Core Items. **

Cohort B: Flexible care adoption rate at Cycle 6.

*Measurement Method:** The MDASI assesses symptom severity and interference. For 6 items (general activity, walking, work, mood, relations, enjoyment of life), patients rate interference over the last 7 days on a 0-10 scale.

7.2 Secondary & Exploratory Endpoints

* Change From Baseline in Global Health Status Score/Quality of Life Score (GHS/QoL) From the European Organisation for Research and Treatment of Cancer (EORTC) Item Library 6 (IL6) GHS/QoL. *

Change From Baseline in the EuroQol 5-Dimension, 5-Level Questionnaire (EQ-5D-5L) Visual Analogue Scale (VAS) Instrument.

* Change From Baseline in Mean Symptom Severity Score From the MDASI Core Items. * Safety, adverse events, and exploratory analysis of Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Proactive digital tracking via the Roche DPM Module, allowing patients to self-report side effects and symptoms remotely. Control arm patients report symptoms per normal practice.

Serious Adverse Events (SAEs): Captured routinely through clinical site assessments and triggered alerts from the DPM algorithm.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) for primary clinical outcomes and PROs; Safety Set for all treatment-emergent AEs.

Sample Size Justification: Original enrollment targets provided adequate power to detect statistically significant differences in the MDASI Total Symptom Interference Score and evaluate feasibility thresholds (study terminated early at 49 targeted enrollment).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted according to the Declaration of Helsinki and/or the applicable laws and regulations of the participating countries. First Ethics Committee approval was obtained in Spain in October 2022.

Data Auditing: App utilization, flexible care adherence rates, and continuous PRO compliance are electronically captured and audited via the Kaiku Health DPM platform.

11. Study Identifiers & Governance

PostingID: 154613381707382742

Primary ID: M042720 / 2023-504342-55-00

Registry Number: NCT05694013

Sponsor/Lead Organization: F. Hoffmann-La Roche Ltd

Study Title: ORIGAMA

Official Regulatory Title: Interventional platform study investigating the impact of digital health solutions on health outcomes and health-care resource utilization in participants receiving systemic treatment in clinical practice

12. Clinical Framework (Oncology & Digital Health Specific)

Disease Areas: Cancer / Malignant Neoplasms

Condition Definition: A neoplasm composed of atypical neoplastic, often pleomorphic cells that invade other tissues. Malignant neoplasms often metastasize to distant anatomic sites and may recur after excision. The most common malignant neoplasms are carcinomas, Hodgkin and non-Hodgkin lymphomas, leukemias, melanomas, and sarcomas.

Ontology Codes:

MESH: D009369

HPO: HP:0002664

SNOMED: 1240414004

Diagnosis Threshold: *Confirmed diagnosis qualifying for Atezolizumab systemic therapy*

Medical Methodology: *Systemic immunotherapy (Atezolizumab IV/SC) supplemented with a Digital Patient Monitoring (DPM) app*

Optimization Target: * Reduction in patient symptom interference and successful transition to flexible (at-home) oncology care

13. Study Procedures & Methodology

Management Pathway: Clinical and flexible at-home oncology administration

Education Protocol (HCP): Administration of SC Atezolizumab in an at-home setting and DPM dashboard monitoring

Education Protocol (Patient): Orientation on the Roche DPM Module (via Kaiku Health) for logging MDASI symptom scores

Quality Audit Mechanism: Digital platform metrics detailing app-usage adherence and health-care resource utilization

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression (Cohort A) or completion of 16 treatment cycles (Cohort B) - study terminated early.

Onsite Visit Cadence: Cohort B utilizes a hospital setting for the first 3 cycles, followed by 13 cycles of flexible at-home care

Remote Monitoring Frequency: Continuous digital tracking via the DPM tool

Checklist Review Interval: Weekly MDASI assessments (last 7 days interference); QoL reporting every 6 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				